

COOPER WHATMORE

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EDUCATION

Colorado School of Mines — Golden, CO
B.S. Engineering Physics | Quantum Engineering Hardware

Dec 2027

TECHNICAL SKILLS

Programming: Python, C++, Mathematica, MATLAB, Quantum Algorithm Strut
Engineering & CAD: Fusion 360, Onshape, circuit design, prototyping, data acquisition
Laboratory: Optical alignment/calibration, high-vacuum systems, cleanroom sample preparation, thermal and ion-beam deposition, lithography
Tools: LabVIEW, Git/GitHub, LaTeX

ENGINEERING & RESEARCH EXPERIENCE

Undergraduate Laboratory Assistant — Quantum Computing Laboratory
Colorado School of Mines — Golden, CO

June 2026-Present

- Support experimental quantum-computing research through laboratory systems management, experimental setup, and data collection.
- Develop Python tools for collecting, processing, visualizing, and fitting experimental data.
- Assist with troubleshooting laboratory instrumentation, hardware interfaces, and experimental control systems.

Engineering / Quantum Technology Intern — Inflection

May 2024-August 2024

Inflection Boulder, CO Lab

- Supported engineering and research activities associated with quantum technology systems and experimental hardware.
- Optics alignment and test environment verification

ENGINEERING PROJECTS

Climbing Fall Dynamics & Accelerometer System
Python | Arduino | ADXL337 | Electronics

- Designing a compact wearable sensing system for measuring acceleration during climbing movement and falls.
- Built real-time serial data acquisition and Python-based visualization/event-analysis tools.

Mechanical Ascender / Descender Concept
Fusion 360 | Mechanical Design

- Developing a rope-management concept combining automatic upward rope feeding with an integrated lowering mechanism.
- Exploring cam geometry, rope path, friction, manufacturability, and compatibility with ~9.6 mm climbing rope.

LEADERSHIP & ADDITIONAL EXPERIENCE

Youth Competition Team Coach — Movement Climbing, Yoga & Fitness

Feb. 2025–Present

Develop and lead structured practices for youth athletes, emphasizing technique, problem solving, safety, and individualized feedback.

Mines Quantum Engineering Organization

Aug 2024 - Present

- Participate in technical events and discussions spanning quantum computing, sensing, and hardware.
- Collaborate with students interested in quantum computing, sensing, hardware, and related technologies.

Society of Physics Students

Aug 2023 - Present

- Participate in technical events and discussions spanning quantum computing, sensing, and hardware.